

NATURAL LANGUAGE PROCESSING APPROACH FOR TRADEMARK
DISTINCTIVENESS CLASSIFICATION

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ABSTRACT

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Trademark law protects marks so that firms can signal the qualities of their products to consumers. Under U.S. trademark doctrine, courts assess whether a mark qualifies for protection by locating it on the Abercrombie spectrum, which classifies marks as fanciful, arbitrary, or suggestive (inherently distinctive) or descriptive and generic (not inherently distinctive). This paper examines whether that classification can be automated using modern natural language processing. We fine-tune ModernBERT on approximately 1.5 million USPTO trademark applications (2012–2019) for binary distinctiveness classification, comparing against DistilBERT and RoBERTa baselines from previous work. ModernBERT yields 91% accuracy versus 89% for baselines. SHAP-based feature attribution confirms that the model’s decisions track legally coherent patterns. Trademark name and goods/services description dominate, and their interaction mirrors the contextual reasoning central to Abercrombie doctrine. Building on this validation, we introduce a confidence-aware triage workflow in which high-confidence predictions are routed for automated disposition while intermediate cases are flagged for human review; error analysis of the high-confidence tails reveals that a substantial fraction of apparent false positives reflect label noise rather than genuine model failure. Together, these results position the system as a viable decision-support tool for trademark examination.

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CHAPTER 1 INTRODUCTION

1.1 Small figures

CHAPTER 2 LITERATURE REVIEW

CHAPTER 3 METHODOLOGY

This study applies ModernBERT to the binary trademark distinctiveness classification task using large-scale USPTO filing data. The model is trained on the full USPTO dataset and evaluated on both the full 2019 test set and a targeted Nice Class 28 (NC28) dataset selected for its relevance to sports and recreation law practice. Evaluation combines standard classification metrics, feature-level SHAP explainability, and a confidence-stratified error analysis designed to assess the viability of a confidence-aware triage workflow.

3.1 Dataset Construction and Input Features

3.1.1 Dataset and Labeling

The training data is drawn from the USPTO Trademark Case Files dataset, supplemented with the full text of approximately 2.2 million office actions linked to roughly 1.5 million applications filed between 2012 and 2019. The dataset construction and labeling procedure follow Adarsh et al. (2024).

The binary classification target is constructed from multiple administrative signals. Applications registered on the Principal Register without a Section 2(f) acquired distinctiveness claim are labeled as inherently distinctive (class 1). Applications refused on descriptiveness or genericness grounds, or registered on the Supplemental Register, are labeled non-distinctive (class 0). Section 2(f) applications are excluded to avoid conflating inherent and acquired distinctiveness. Image-only marks and marks of five or more words are also excluded, representing approximately 20% and 6% of filings respectively. A temporal train/validation/test split is applied: 2012–2017 training, 2018 validation, 2019 held-out test. This year-based split prevents data leakage and reflects the prospective deployment context.

Table 3.1 Label construction rules for the binary distinctiveness target.

Condition	Label	Class
Registered on Principal Register, no Section 2(f) claim	Inherently distinctive	1
Refused on descriptiveness or genericness grounds	Non-distinctive	0
Registered on Supplemental Register	Non-distinctive	0

3.1.2 NC28 Dataset

In addition to this full-dataset evaluation, this study conducts a targeted analysis on all Nice Class 28 (sports equipment and sporting goods) applications in the 2019 test set, which is referred to throughout as the NC28 dataset. NC28 contains 10,251 observations with a base rate of approximately 89.7% inherently distinctive marks. This targeted analysis is audience-driven as the primary venue for this work is the Sports and Recreation Law Association conference. NC28 is not a substitute for the full evaluation; it is an additional layer of analysis applied after full-dataset performance is established.

3.1.3 Input Feature Construction

Each trademark application is represented as a single concatenated text string constructed from eight structured fields: (1) the mark text; (2) the goods and services description; (3) the Nice class description; (4) the pseudo-mark, a standardized representation that strips stylistic elements; (5) a WordNet dictionary presence flag; (6) translation status; (7) the numeric Nice class category; and (8) mark character length. Fields are joined with delimiter tokens to preserve structural boundaries. This field-delimited structure is integral to the feature-level SHAP analysis.

3.2 Model Selection and Architecture

This study adopts ModernBERT-base as its primary classification model. Trained on 2 trillion tokens with a native 8,192 token sequence length, ModernBERT achieves state-of-the-art results across diverse classification tasks and retrieval benchmarks. [!!arXiv](#) Three features are particularly relevant to the trademark classification task.

First, ModernBERT replaces absolute positional embeddings with rotary positional embeddings (RoPE). RoPE encodes position as a function of relative token distance rather than absolute index, which produces more stable representations for variable-length inputs. [!!arXiv](#) Trademark applications vary substantially in length, and relative positional encoding handles this variability more robustly than fixed absolute embeddings.

Second, the context window extends to 8,192 tokens, compared to 512 for both

DistilBERT and RoBERTa. USPTO goods-and-services descriptions can be lengthy, particularly for multi-class applications. Truncating at 512 tokens risks discarding legal description content that the model would use to assess the mark–goods relationship, which is a central factor in the Abercrombie analysis.

Third, ModernBERT was pretrained on approximately two trillion tokens spanning web text, scientific literature, and code. This is substantially broader than the books-and-Wikipedia corpora used by its predecessors. The wider pretraining distribution provides richer coverage of the heterogeneous vocabulary found across Nice classes, where goods-and-services descriptions range from technical manufacturing language to consumer product terminology.

The DistilBERT and RoBERTa results from Adarsh et al. (2024) serve as baseline comparisons. No modifications were made to those models’ configurations; they are referenced as reported. Including them without retraining ensures that performance differences are attributable to the model architecture rather than changes in data, labeling, or training procedure. Table 3.2 summarizes the key architectural differences across all three models.

3.3 Training Procedure

ModernBERT-base is fine-tuned for binary sequence classification using a standard linear classification head appended to the [CLS] token representation, with sigmoid activation producing a probability for the inherently distinctive class.

Training uses a weighted binary cross-entropy loss function. Class weights are computed from the training label distribution to counteract majority-class bias: approximately 84% of training marks are labeled inherently distinctive, so without reweighting the model would default to the majority class. The weighting scheme penalizes misclassification of non-distinctive marks proportionally more, pushing the model toward balanced decision boundaries.

The model is trained for two epochs with a learning rate of 5×10^{-6} and a per-device batch size of 16 across the full 1.5 million-application training set. All experiments are conducted on GPU infrastructure at the Ohio Supercomputer Center using Slurm-scheduled Apptainer containers for reproducibility.

Table 3.2 Architectural comparison of DistilBERT, RoBERTa, and ModernBERT-base.

Aspect	DistilBERT	RoBERTa	ModernBERT
Architecture	2019-era distilled Transformer; knowledge-distilled from BERT-base	Standard 2018-era Transformer encoder	Modernized encoder with LLM-inspired refinements (2024)
Positional Encoding	Absolute	Absolute	Rotary (RoPE)
Attention	Full global in all layers	Full global in all layers	Alternating global and local
Max Context	512 tokens	512 tokens	8,192 tokens
Pretraining Data	Wikipedia + Toronto BookCorpus	Books + Wikipedia	~2T tokens: web, scientific text, code
Parameters	~66M	~125M	~149M
Efficiency	~60% faster than BERT-base via layer halving and distillation	Competitive on short fixed-length inputs	Optimized for variable-length and long-sequence inputs
Benchmark Trends	~97% of BERT-base GLUE at lower compute	Strong NLU performance	Comparable or improved NLU; stronger retrieval and long-context

No post-hoc probability calibration is applied. The confidence-stratified error analysis in Chapter 7 treats raw predicted probabilities as interpretable confidence signals, an assumption informed by the prior study’s finding that both DistilBERT and RoBERTa were well-calibrated without correction !!! . Whether ModernBERT exhibits similar calibration behavior is examined in the next Chapter.

3.4 Evaluation Framework

3.4.1 Standard classification metrics

Performance is reported using accuracy, precision, recall, F1 score, and ROC–AUC at a fixed 0.5 probability threshold. Both macro-averaged and weighted F1 are reported to make the trade-off between aggregate accuracy and minority-class detection explicit. Balanced accuracy is

included as an imbalance-robust summary metric. Metrics are computed on the full 2019 test set for comparability with the baselines, with additional reporting on the NC28 dataset

3.4.2 Feature-level explainability

SHAP !!!(Lundberg & Lee, 2017) analysis is conducted on the NC28 dataset to assess whether the model’s learned patterns are consistent with Abercrombie doctrine. This study assigns attribution values at the feature-field level as each of the eight structured input fields receives a single SHAP value per prediction. This design choice aligns the explainability output with how trademark examiners reason: as discrete inputs such as the mark text, goods description, and dictionary status, not subword tokens. A zero baseline is used as the reference distribution. Global summary plots identify which fields most consistently drive predictions; local waterfall plots decompose individual cases.

3.4.3 Confidence-stratified error analysis

Aggregate accuracy does not distinguish between errors made at high confidence(where the system would act without prompting review) and errors in the uncertain middle range (where the system would correctly defer to a human). A model that makes errors primarily in uncertain regions is a useful triage instrument; one that produces high-confidence errors is not.

Three confidence-defined subsets of the NC28 dataset are examined. High-confidence false positives are cases where the predicted probability exceeds 0.9 but the true label is non-distinctive; high-confidence false negatives are cases where the predicted probability falls below 0.1 but the true label is distinctive. These thresholds define the operating range where a triage system would most likely route cases to automated outcomes. Predictions above 0.9 are redirected toward a publication recommendation, while that below 0.1 are redirected toward refusal without examiner review. A third subset examines borderline predictions in the 0.4–0.6 band, where model uncertainty is highest and the triage workflow would appropriately direct cases to human review. For each subset, error counts and proportions are reported and the associated mark text and goods descriptions are examined for recurring linguistic patterns.

Accuracy across confidence bins is also reported, characterizing the distribution that directly informs the triage threshold recommendations in the Discussion.

CHAPTER 4 RESULTS

This chapter reports classification performance, feature-level interpretability analysis, and targeted error investigation for ModernBERT. We assess classification performance using standard metrics (accuracy, precision, recall, F1, ROC-AUC) computed at a fixed 0.5 decision threshold, alongside confidence-stratified accuracy analysis to assess whether prediction confidence corresponds to classification reliability. For comparative context, we reference DistilBERT and RoBERTa performance from Adarsh et al. (2024) where relevant. However, the primary focus remains on ModernBERT as the current work’s principal contribution.

4.1 Overall Metrics

Table ?? reports classification metrics for ModernBERT alongside DistilBERT and RoBERTa baselines, all evaluated on the NC28 dataset using a fixed 0.5 decision threshold. ModernBERT achieves 91% accuracy, outperforming both baselines by approximately 2 percentage points. Weighted precision and F1 scores similarly favor ModernBERT, reflecting improved performance on the majority-class distinctive marks. However, balanced accuracy, which adjusts for class imbalance by averaging recall across both classes, remains modest at 0.59, revealing persistent difficulty in minority-class detection. ROC-AUC values cluster tightly in the 0.76–0.78 range across all three architectures, suggesting that discriminative power plateaus regardless of model sophistication. The remainder of this section focuses exclusively on ModernBERT.

Table 4.1 Comparison of Performance Metrics across BERT Model Variants

Metric	DistilBERT	RoBERTa	ModernBERT
Accuracy	0.86	0.86	0.88
Balanced Accuracy	0.60	0.63	0.65
Precision (Weighted)	0.83	0.85	0.87
F1 (Weighted)	0.82	0.84	0.86
ROC-AUC	0.76	0.77	0.77

4.2 Confusion Matrix and Class Imbalance

The confusion matrix (Figure 4.1) exposes the consequences of class imbalance. ModernBERT achieves 99% recall on distinctive marks but substantially lower specificity on non-distinctive marks, systematically biasing borderline cases toward the majority class. The gap between weighted F1 (0.88) and macro F1 (0.62) quantifies this asymmetry: distribution-weighted metrics overstate effective performance relative to metrics that treat both classes equally. This pattern is characteristic of imbalanced classification tasks and reflects the model's learned preference for the dominant label when faced with ambiguous inputs.

Confusion Matrix (threshold = 0.5)

True Label	True Non-Distinctive	189	870
	True Distinctive	90	9,102
		Prediction Non-Distinctive	Prediction Distinctive
		Predicted Label	

Figure 4.1 Confusion Matrix.

4.3 Class Separability and ROC-AUC

ModernBERT achieves a ROC-AUC of 0.768, indicating moderate separability between distinctive and non-distinctive marks. The relatively flat ROC curve beyond FPR = 0.3 signals limited room for threshold adjustment to improve precision without sacrificing recall. An AUC below 0.80 typically indicates substantial overlap in predicted probability distributions, though whether this reflects model limitations or inherent task ambiguity remains unclear. The narrow AUC range (0.76–0.78) across all three architectures suggests performance is constrained by factors beyond model design.

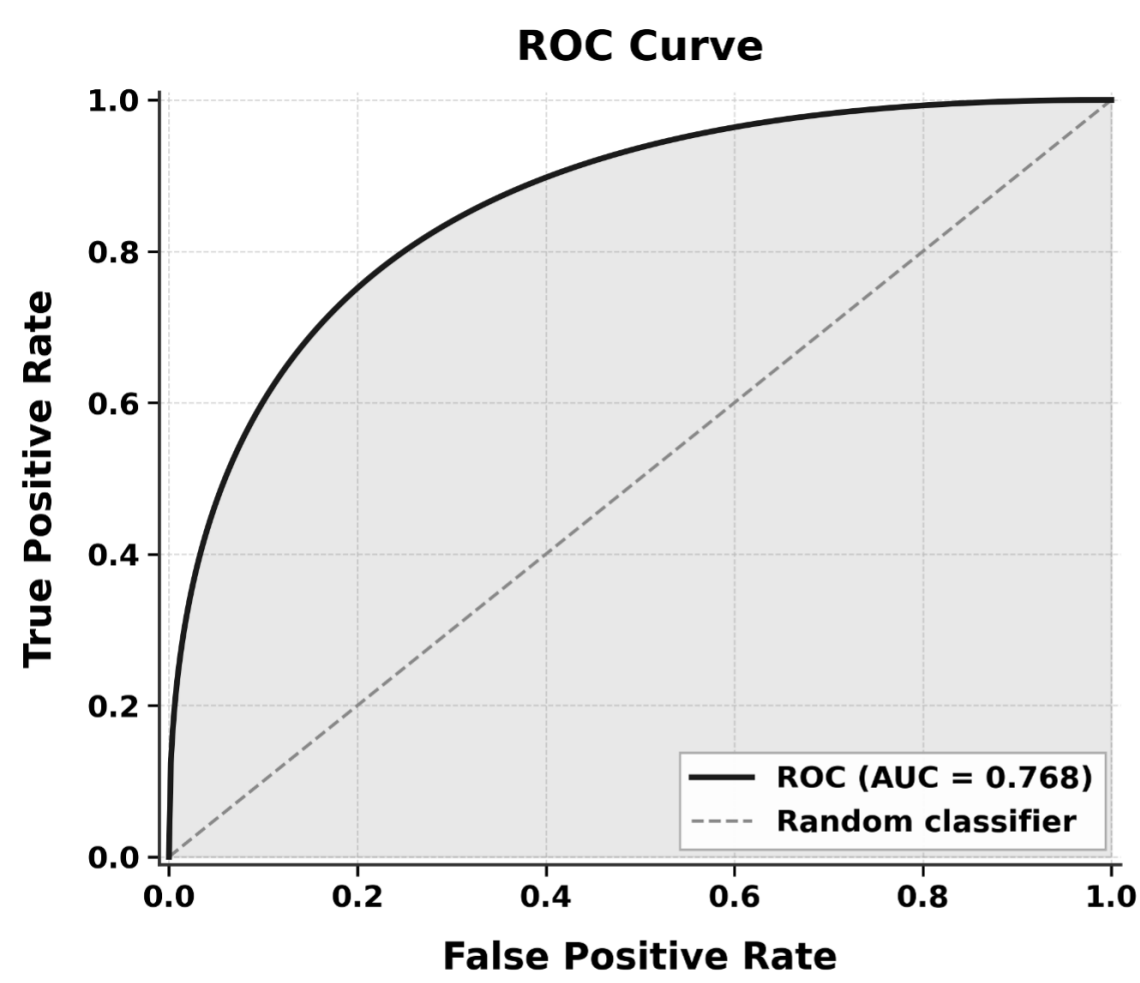


Figure 4.2 ROC curve showing moderate class separability with AUC = 0.768.

4.4 Probability Distribution and Confidence Concentration

Figure 4.3 displays the distribution of ModernBERT’s predicted probabilities. The distribution is heavily right-skewed, with the majority of predictions concentrated above 0.9 for the distinctive class. This reflects both the dataset’s 89.7% base rate and the model’s strong majority-class recall (0.99). Notably, the intermediate range (0.4–0.6) contains relatively few predictions, indicating the model rarely expresses uncertainty.

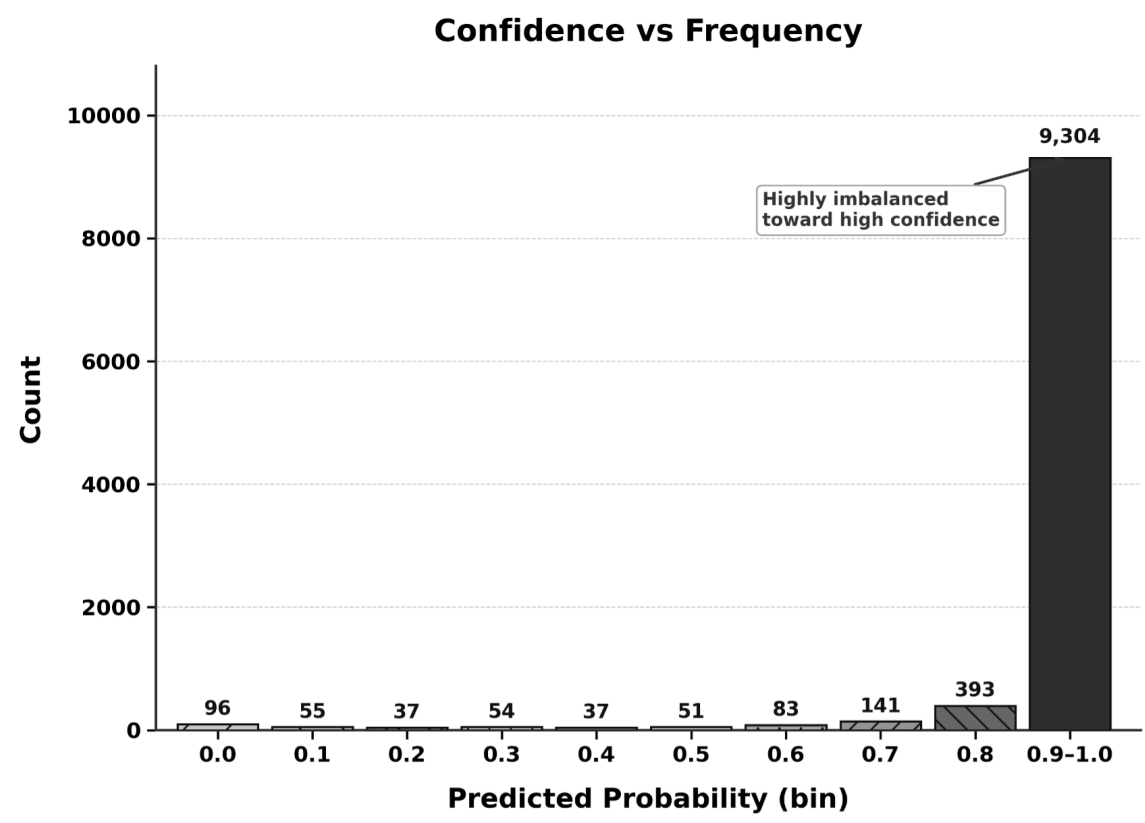


Figure 4.3 Distribution of predicted probabilities on NC28.

4.5 Confidence and Accuracy

To assess the relationship between prediction confidence and classification accuracy, we bin observations by predicted probability and compute empirical accuracy within each bin. Figure 4.4 reveals a U-shaped relationship between confidence and accuracy: predictions at the extremes ($p < 0.2$ or $p > 0.8$) correspond to higher accuracy, while mid-range predictions ($0.4 < p < 0.6$) exhibit elevated error rates. High-confidence predictions tend to involve clearer cases, including

either marks that are fanciful (e.g., SFUMOC) or transparently descriptive (e.g., AIR DRY CLAY). Borderline probabilities correspond to marks at contested boundaries on the Abercrombie spectrum where classification difficulty is inherent to the legal doctrine itself rather than a model deficiency.

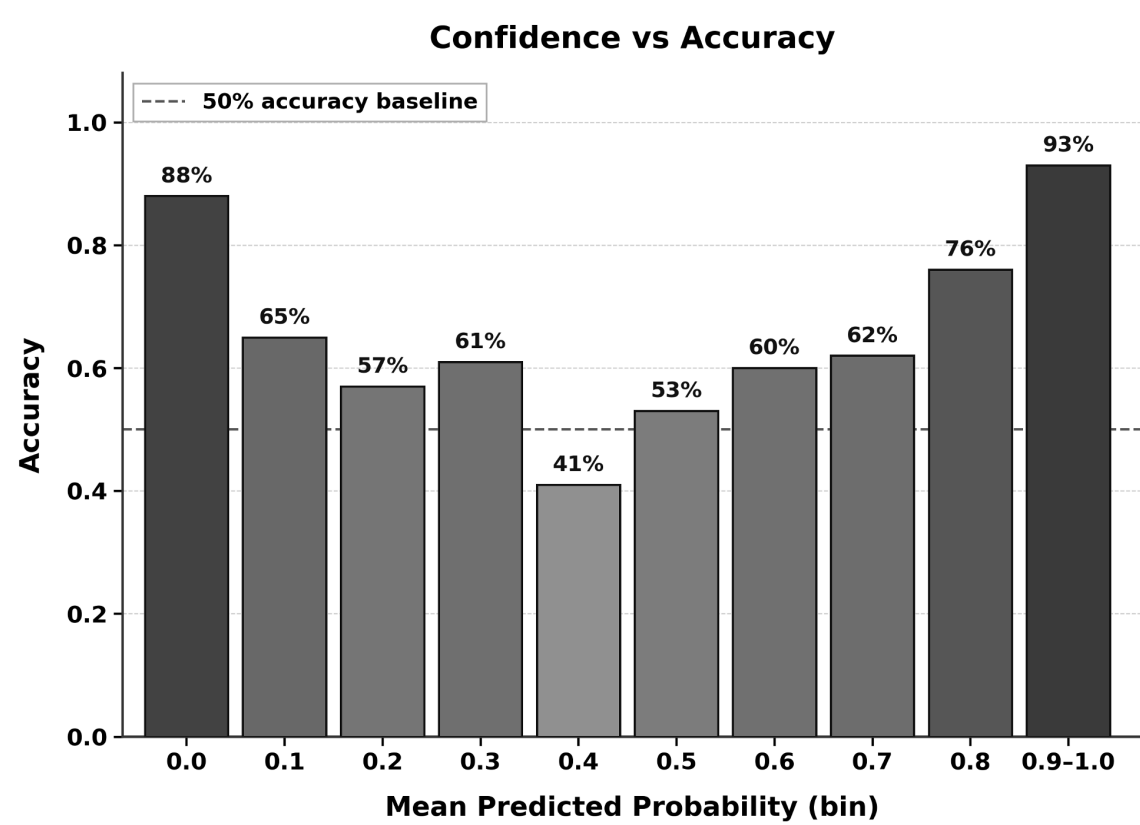


Figure 4.4 Empirical accuracy by predicted probability bin.

CHAPTER 5 MODEL INTERPRETABILITY

To understand which input features drive predictions, we apply SHAP (Shapley Additive Explanations) at the feature-field level. SHAP values quantify each field’s contribution to the predicted probability relative to a fully masked baseline, enabling both global importance rankings across the dataset and local explanations for individual predictions.

5.1 Global Feature Importance

Figure 5.1 ranks features by their mean absolute SHAP value, measuring average influence on model output. Trademark name and its description dominate, each contributing approximately an order of magnitude more than other inputs. This aligns with the Abercrombie framework’s emphasis on the relationship between a mark’s textual content and its associated goods or services. Nice class description, pseudo-mark, and WordNet flag contribute moderately, while translation status, Nice category identifier, and mark length exert minimal average influence. The hierarchy is unsurprising: distinctiveness determinations hinge on semantic relationships between words and products, which mark text and description capture directly.

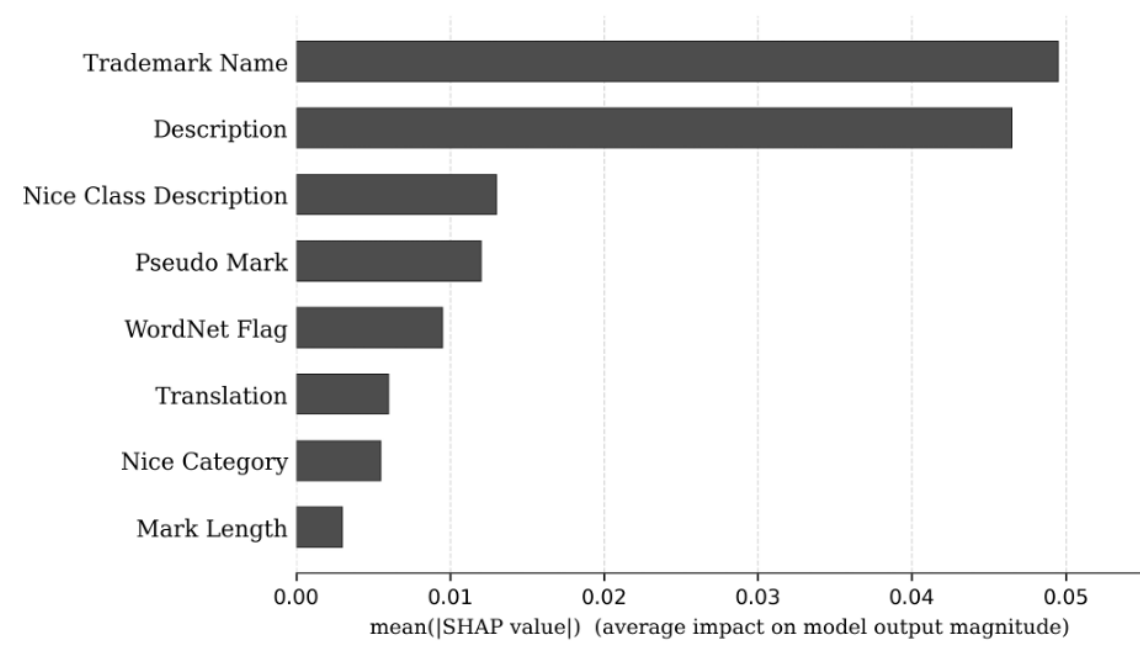


Figure 5.1 Empirical accuracy by predicted probability bin.

5.2 Local SHAP Explanations

To illustrate feature contributions at the individual prediction level, we examine two contrasting cases: SFUMOC, classified as distinctive with high confidence ($p = 0.997$), and AIR DRY CLAY, classified as non-distinctive with high confidence ($p = 0.102$).

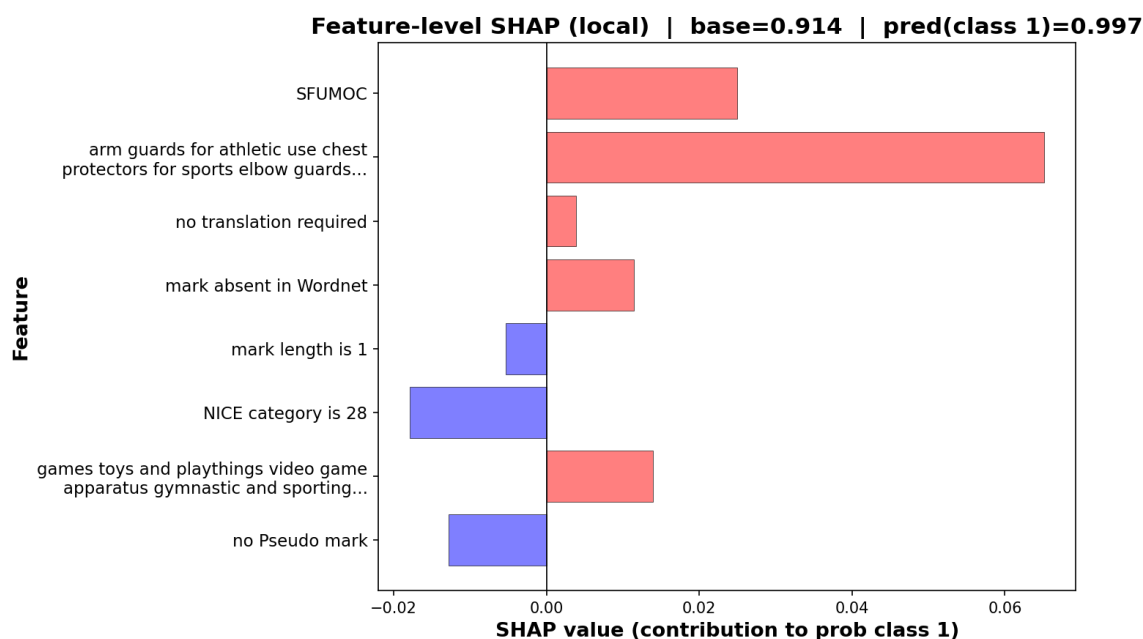


Figure 5.2 Local SHAP decomposition for SFUMOC (predicted probability = 0.997 for distinctive class).

Figure 5.2 displays the SHAP waterfall plot for SFUMOC, predicted as distinctive with probability 0.997. Starting from the dataset base rate of 0.914, the model incrementally adjusts its prediction based on feature contributions. The goods-and-services description (“arm guards for athletic use chest protectors for sports elbow guards for athletic use hand pads for athletic use knee guards for athletic use leg guards for athletic use pet toys shin pads for athletic use waist protectors for athletic use wrist guards for athletic use arm protectors for athletic use the wording sfumoc has no meaning in a foreign language”) contributes the largest positive shift (+0.06), pushing the probability upward. The trademark name itself (“SFUMOC”) adds an additional +0.025. Secondary features reinforce the distinctive prediction: the mark’s absence from WordNet (+0.01) and the Nice class description (+0.015) both signal fanciful or coined character.

Two features exert minor negative pressure: Nice category 28 and the absence of a pseudo-mark. However, these are outweighed by the positive contributors. The cumulative effect moves the prediction from 0.914 to 0.997, reflecting high confidence that SFUMOC is inherently distinctive. The reliance on mark text and goods-description aligns with Abercrombie doctrine: a coined term applied to athletic equipment shows no semantic relationship between the mark and the product, the hallmark of fanciful distinctiveness.

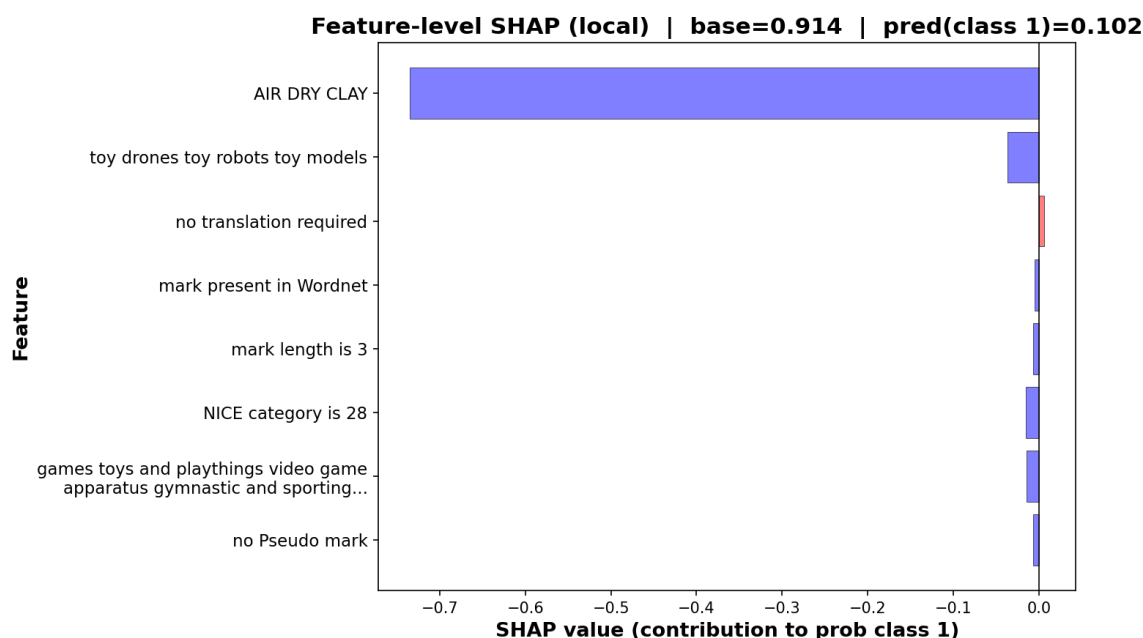


Figure 5.3 Local SHAP decomposition for AIR DRY CLAY (predicted probability = 0.102 for distinctive class).

Figure 9 presents the opposing case. AIR DRY CLAY receives a predicted probability of 0.102, driven almost entirely by the trademark name feature. Specifically, it contributes -0.7 to the prediction, accounting for approximately 86% of the total shift from the base rate. The three-word phrase comprises common English terms ("air," "dry," "clay") that directly describe a product attribute: clay that air-dries. This transparent descriptiveness triggers a strong negative signal. The goods-and-services description ("toy drones toy robots toy models") and other metadata features contribute minor additional negative pressure, but the mark text alone determines the outcome. Unlike SFUMOC, where multiple features converged to reinforce distinctiveness, AIR DRY CLAY's classification rests on a single dominant signal.

CHAPTER 6 ERROR ANALYSIS

This chapter reports a targeted error investigation for ModernBERT on the NC28 dataset. To assess the nature of misclassifications, we stratify errors by prediction confidence and examine three categories. They are high-confidence false positives (predicted distinctive with $p > 0.9$ but labeled non-distinctive, $n = 664$), high-confidence false negatives (predicted non-distinctive with $p < 0.1$ but labeled distinctive, $n = 12$), and borderline errors ($0.4 < p < 0.6$, prediction disagrees with label, $n = 46$). This confidence-stratified approach isolates the most consequential errors—those where the model expresses strong certainty but disagrees with ground-truth labels.

6.1 High-Confidence False Positives

Among the 664 high-confidence false positives, we examine current USPTO registration status to determine whether the model’s prediction or the ground-truth label better reflects the mark’s actual trajectory. Table 6.1 reports the breakdown.

Table 6.1 USPTO Disposition of High-Confidence False Positives ($n = 664$)

Status	Count	Percentage
Abandoned	468	71.0%
Registered/Maintained	166	25.2%
Cancelled	7	1.1%
Suspended/Other	18	2.7%

The 71% abandonment rate is striking. Abandoned applications never receive a final merits determination on distinctiveness. Many applicants may abandon for procedural reasons (failure to respond to office actions), strategic reasons (cost-benefit analysis), or reasons unrelated to the mark’s distinctiveness (business pivot, trademark no longer needed). In these cases, the ground-truth label reflects a preliminary examiner assessment that was never adjudicated to completion. The model’s high-confidence distinctive prediction may be substantively correct for a substantial share of these 468 marks, with the apparent “error” attributable to label noise rather than classification failure. The 166 marks (25.2%) that reached registration represent a simpler case: these applications were filed before the dataset construction cutoff but registered afterward.

The ground-truth label captured their initial filing status while the model’s prediction proved consistent with their eventual outcome. Taken together, the disposition analysis suggests that a significant fraction of high-confidence false positives are mislabeled in the training data rather than genuine model errors.

6.2 High-Confidence False Negatives

High-confidence false negatives are rare: only 12 observations receive predicted probabilities below 0.1. Manual inspection suggests these errors trace to labeling inconsistencies rather than systematic model failures. The low amount of high-confidence false negatives aligns with the confusion matrix findings: the model rarely assigns high confidence to the non-distinctive class, consistent with its strong majority-class recall and low specificity. The model’s primary failure mode is majority-class bias rather than confidently incorrect minority-class predictions.

6.3 Borderline Errors and Legal Ambiguity

The 46 borderline errors occupy the predicted probability range 0.4–0.6, where model uncertainty is highest. To illustrate the nature of these cases, we examine PARTYBALLOONS, which receives a predicted probability of 0.575 despite a non-distinctive ground-truth label.

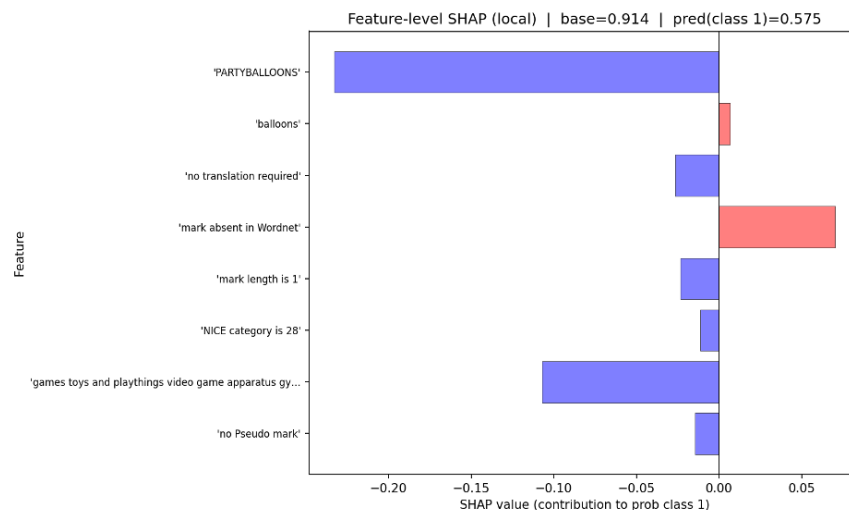


Figure 6.1 Local SHAP decomposition for PARTYBALLOONS (predicted probability = 0.575).

Figure 6.1 displays the SHAP waterfall plot for PARTYBALLOONS, which receives a

predicted probability of 0.575 despite a non-distinctive ground-truth label. Starting from the base rate of 0.914, the mark text "PARTYBALLOONS" contributes the largest negative shift (-0.18), pulling the prediction downward. The Nice Class 28 description adds an additional -0.13, reinforcing the non-distinctive signal. Together, these two features account for the bulk of the negative pressure. One of the features pushes back toward distinctive: the mark's absence from WordNet (+0.05) which suggests it is not be a standard dictionary term.

This case illustrates a critical point about borderline predictions. The compound combines two ordinary English words that fairly directly describe a product category. Whether such compounds are merely descriptive or suggestive involves subtle semantic judgments. The 46 borderline cases represent marks where model uncertainty aligns with legal uncertainty. Higher misclassification rates in the intermediate probability range stem from inherent classification difficulty, not model failure.

CHAPTER 7 DISCUSSION AND LIMITATION

7.1 ROC-AUC Performance Ceiling

ModernBERT achieves 91% accuracy on the NC28 evaluation set, yet ROC-AUC remains at 0.768 which is consistent with DistilBERT and RoBERTa baselines. This plateau across architectures is better understood as a property of the task than a limitation of the model. Abercrombie determinations are inherently dependent on judgment as the boundary between suggestive and descriptive marks is contested even among experienced examiners and courts. A model trained on historical USPTO outcomes absorbs that contestation into its uncertainty. Expecting near-perfect classification would misunderstand what the legal determination requires. The more productive question is not whether the model can resolve every case, but whether it can reliably identify the cases it should and should not resolve and the results suggest it can.

7.2 Interpreting Model Confidence

The system's practical value lies not in its binary prediction at a fixed threshold but in the calibrated probability it produces. A calibrated probability is a genuine likelihood estimate. To be specific, a prediction of $p = 0.92$ means the model assigns 92% confidence to a distinctive outcome, not merely that the mark cleared a decision boundary. The U-shaped relationship between confidence and accuracy in Figure 4.4, where predictions at the extremes are substantially more accurate than those in the intermediate range, means that confidence level is itself informative. High-confidence outputs warrant stronger reliance; mid-range outputs reflect genuine ambiguity at the contested regions of the Abercrombie spectrum, and should be treated accordingly.

7.3 The Confidence-Aware Triage Workflow

These properties motivate a three-zone triage workflow for decision support. At the high-confidence distinctive end, predictions can be routed toward expedited processing with low examiner burden, combining with SHAP-based explanations showing which mark features drove the prediction. This provides a transparent and auditable basis for that routing. At the

high-confidence non-distinctive end, predictions flag applications as strong candidates for descriptiveness or genericness refusal, again with feature-level explanations that examiners or attorneys can interrogate. In the intermediate zone, where the model's uncertainty is highest and accuracy lowest, the system defers to human judgment. This workflow is designed for both USPTO examiners seeking consistency and efficiency and outside trademark counsel assessing registration prospects.

7.4 Error Analysis Implications

Inspection of high-confidence false positives reinforces this framing. Of the 664 cases where the model predicted distinctiveness with high confidence but the label indicated otherwise, 71% were subsequently abandoned. An additional 25% were eventually registered after the dataset cutoff, meaning the model's prediction aligned with the ultimate outcome. Apparent errors at the high-confidence tail are artifacts of label construction rather than genuine model failures. This finding strengthens the case for relying on the tails as the most credible zone of the triage workflow. Several constraints on the current system bear noting

7.5 Label Construction and Class Imbalance

The binary classification target collapses three legally distinct categories into a single positive class. Fanciful, arbitrary, and suggestive marks occupy genuinely different positions on the Abercrombie spectrum, and treating them as a single "distinctive" label glosses over distinctions that matter. Coined marks and suggestive ones are not legally the same. That distinction can matter in opposition and infringement proceedings. Collapsing them into a single label was a practical necessity, since reliable fine-grained annotations simply do not exist at scale. The evaluation dataset adds another layer of complexity. NC28 applications carry an 89.7% distinctive base rate, which reflects how USPTO filings actually look in practice, but it also means the model is structurally pushed toward predicting the majority class. The gap between weighted F1 (0.88) and macro F1 (0.62) follows directly from that imbalance. Put plainly, the headline performance numbers are somewhat flattering. The model is less reliable on non-distinctive

marks, and those are precisely the cases where getting it wrong carries the most practical consequence.

7.6 Interpretive Limitations of the Model

SHAP attribution in this study works at the feature-field level, measuring how much each structured input field, like mark text or goods and services description, contributes to the predicted probability. That is enough to show that the model's decisions track legally coherent patterns, which is the main thing the analysis is trying to establish. However, it does not reveal which specific tokens within a field are actually doing the work. A goods and services description might be driving a prediction because of one descriptive word, or because of the overall semantic weight of the whole statement. The current analysis cannot tell the difference between those two scenarios, and that is a real limitation worth acknowledging.

Additionally, the model is trained on historical USPTO examination outcomes, which are themselves the product of human adjudication, and human adjudication is not perfectly consistent. Examiner variance, shifting USPTO guidance, and the inherent subjectivity of Abercrombie determinations all mean the training signal is noisy rather than clean. The model learns what examiners decided during a particular period, not what the doctrine actually requires. This distinction really matters when the system is used to assess marks at the contested boundaries of the spectrum.

CHAPTER 8 CONCLUSION

BIBLIOGRAPHY